

AI-POWERED CYBER THREAT DETECTION



INTRODUCTION

As digital systems grow more interconnected, the risk of cyberattacks continues to rise. Traditional security systems are no longer sufficient to handle the complexity and scale of modern cyber threats. This project focuses on building an intelligent Intrusion Detection System (IDS) using machine learning, trained on the UNSW-NB15 dataset — a comprehensive and realistic dataset that includes various types of network traffic. The goal is to develop a system that can automatically detect and classify cyberattacks in real time, improving security response and reducing false alerts.

OVERVIEW/BACKGROUND INFORMATION

Cybersecurity threats such as malware, DoS attacks, ransomware, and phishing—are growing in both frequency and sophistication, with global damages projected to reach \$10.5 trillion annually by 2025.



Traditional rule-based IDS struggle to detect new threats and generate many false positives.

This project develops a machine learning-based IDS using the UNSW-NB15 dataset to detect, classify, and interpret cyber threats in real time, offering a more adaptive and intelligent defense against evolving attacks."

PROBLEM STATEMENT



Current network security systems face increasing difficulty in detecting cyber threats due to their evolving nature. Key issues include:

- High false negatives lead to undetected threats, causing severe damage and high recovery costs
- Limited adaptability, as rule-based systems fail to detect unknown (zero-day) attacks
- Complex traffic behavior, making it hard to distinguish normal anomalies from malicious actions
- Slow detection times, which give attackers more opportunity to cause harm

OBJECTIVES

Explore the distribution of attack types

Provide clear explanations for model predictions
using LIME

Train machine learning models to classify
the kinds of network traffic

Simulate real-time detection through a stream-based
pipeline

Detect unknown or zero-day attacks using anomaly
detection techniques

Times New Roman

DATA UNDERSTANDING

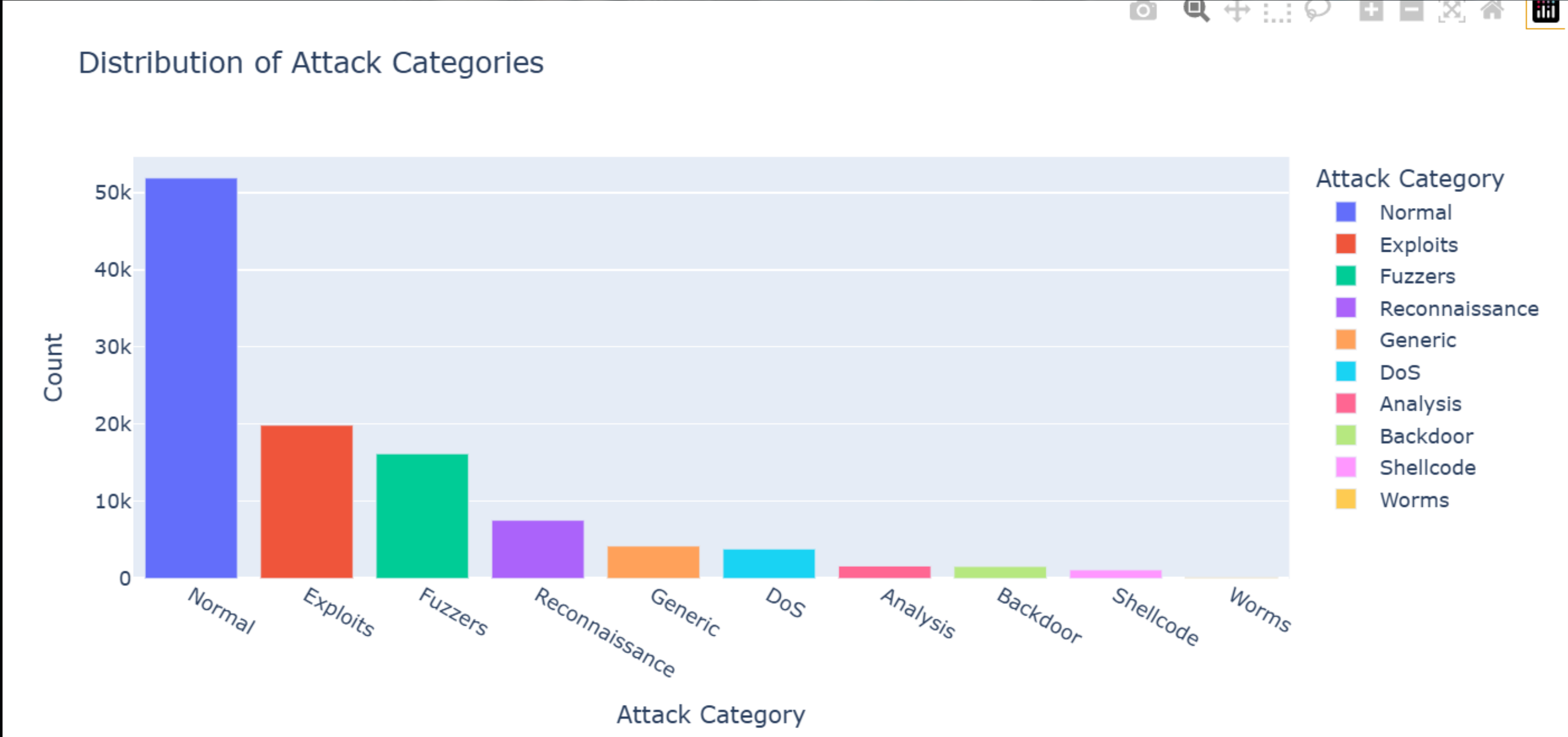


- Source: Australian Centre for Cyber Security (UNSW Canberra)
- Purpose: Simulates real-world benign and 9 types of malicious network traffic
- Collection Tools: IXIA PerfectStorm (traffic), Argus & Bro/Zeek (feature extraction)

Key Stats:

- Total Records: 2.54 million
- Train Set: 175,341 records
- Test Set: 82,332 records
- Features: 49 (including IP metadata, time-based stats, and content-based attributes)
- 9 attack types: DoS, Exploits, Reconnaissance, Shellcode, Fuzzers, Analysis, Backdoors, Generic, Worms

DATA INSIGHTS

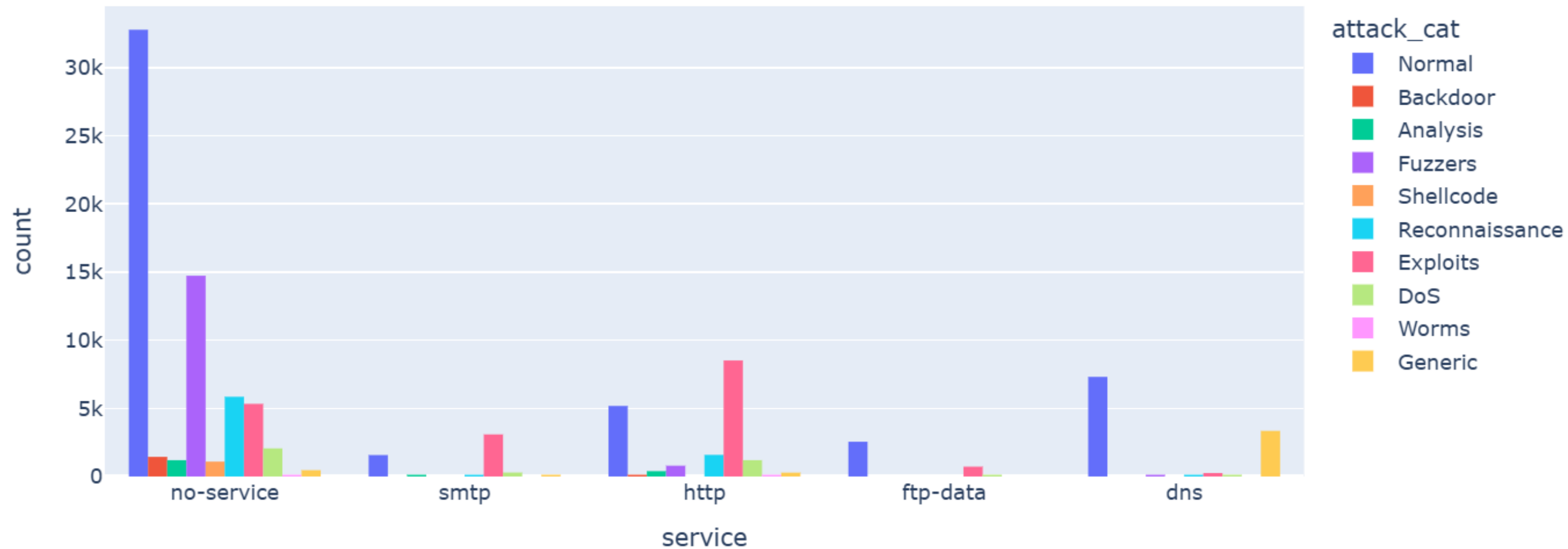


Most Frequent category: Normal

Least Frequent category: Worms

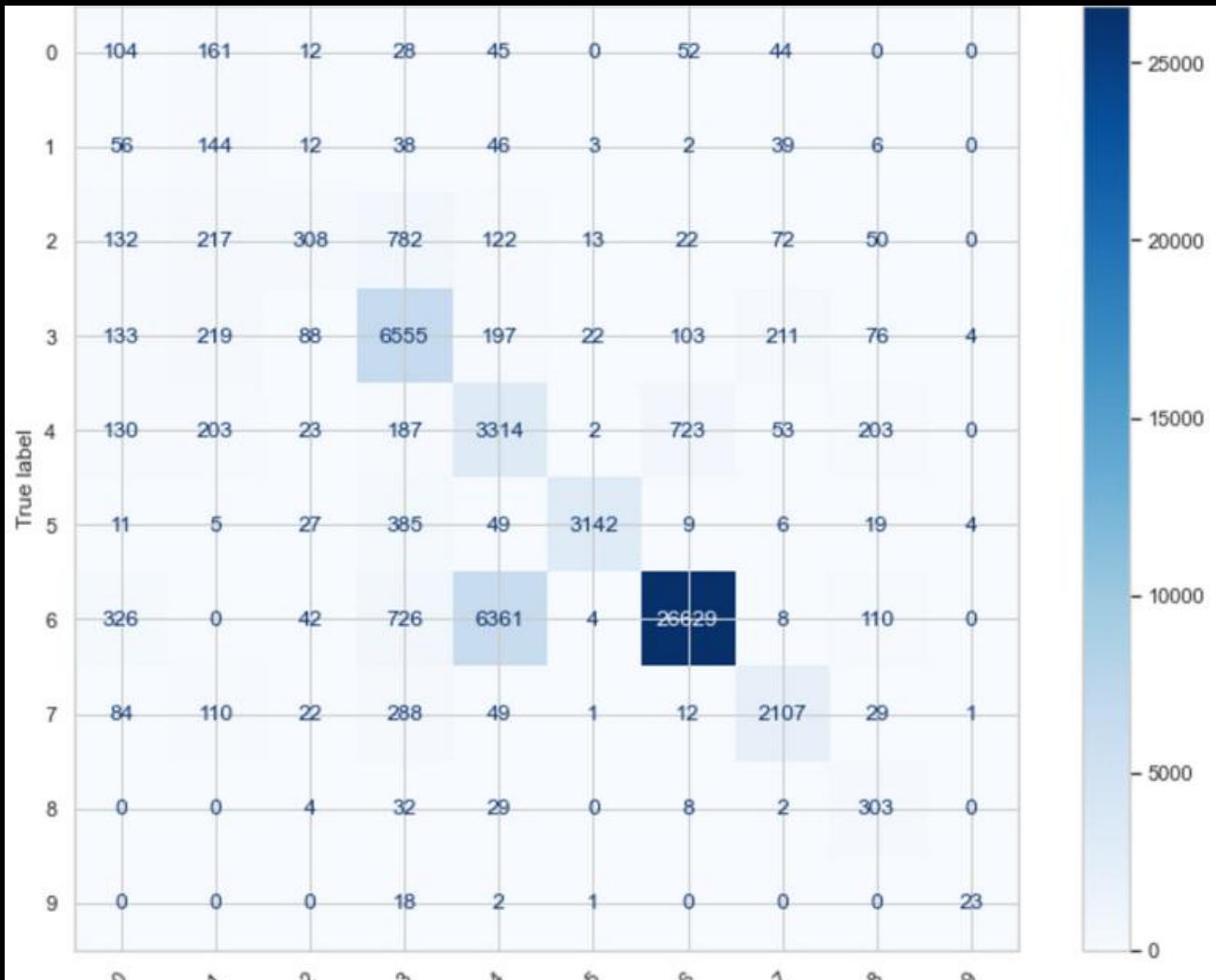
DATA INSIGHTS

Top 5 Services vs Attack Category



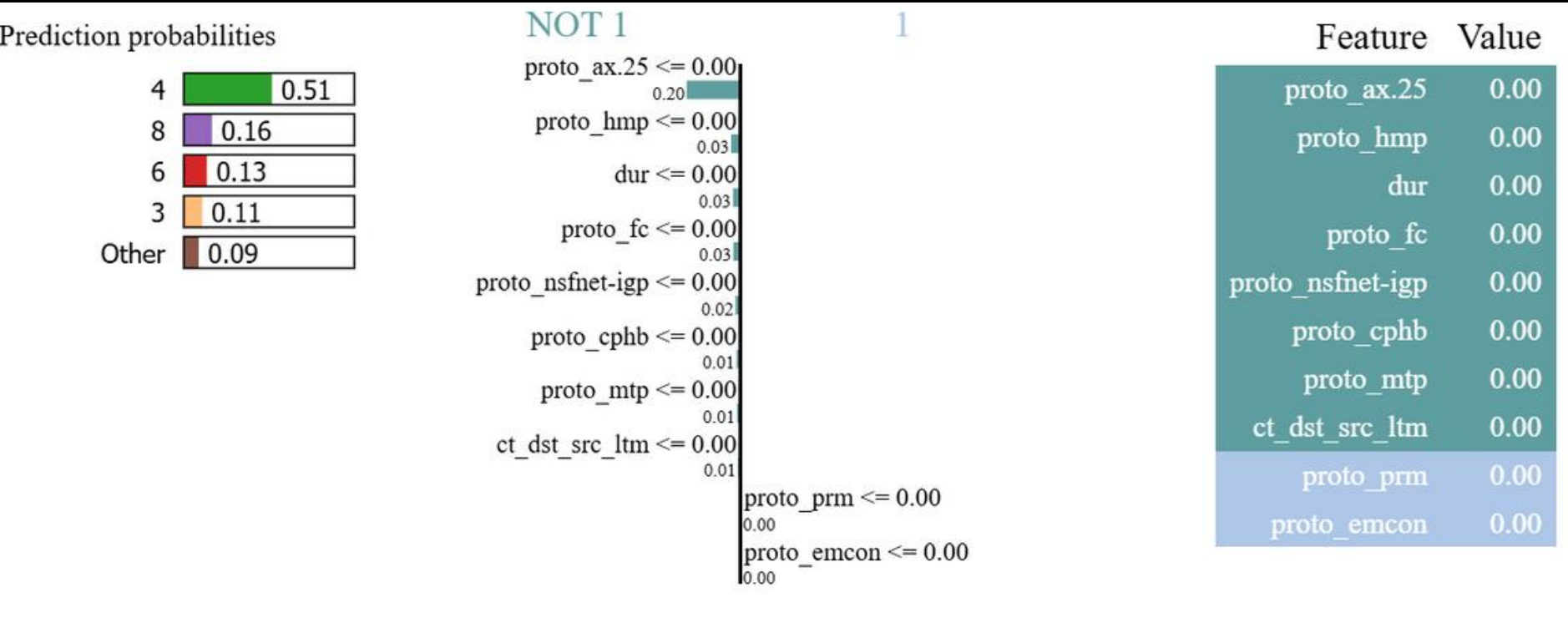
The graph above shows the count of each attack category. It reveals that the most targeted category is 'no_service', which exhibits a range of attack counts from high to low. This is followed by 'http', although its attack counts are relatively lower.

MODELING : XGBOOST PERFORMANCE



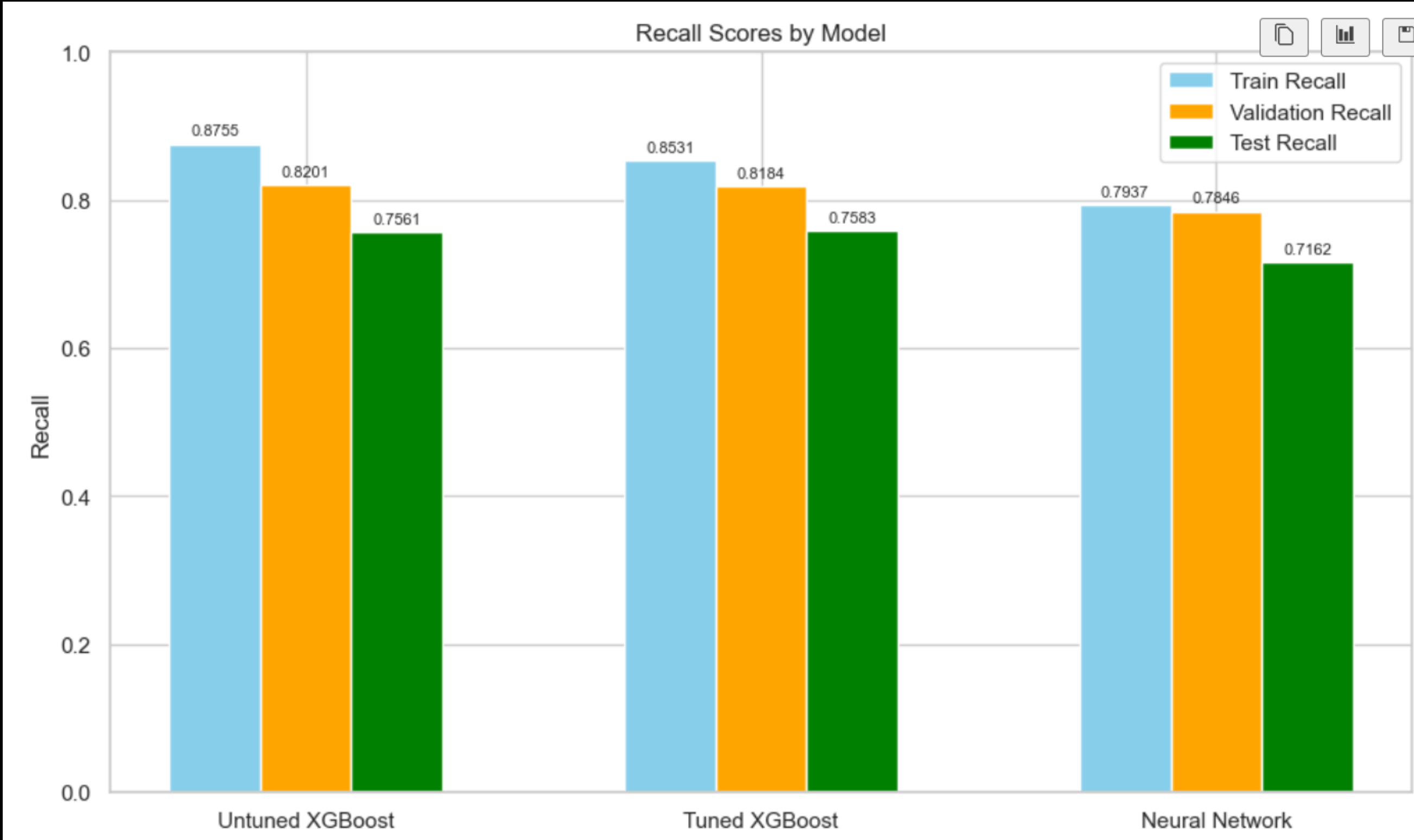
Tuned XGBoost Confusion Matrix

- Strong diagonal dominance → most predictions correct.
- Class 6: Excellent performance, minor confusion with Classes 3 & 4.
- Classes 3 & 4: Moderate confusion with nearby classes.
- Classes 8 & 9: Few samples → class imbalance challenges.



- LIME explained why the model predicted this case (Index 0) as malicious with 100% probability.
- Green bars = features pushing toward 'malicious', red bars = features pushing away (none here).
- Top contributors: ct_state_ttl, state_CON, ct_dst_sport_ltm — all strongly indicated malicious activity

STACKED BAR CHART



Recall scores are highest for Untuned XGBoost (Train: 0.88, Test: 0.76). Tuning slightly improves test recall(0.758 → 0.7583). Neural Network shows lower recall across all splits but more consistent performance.

CONCLUSIONS

- Supervised ML models, especially Tuned XGBoost, outperformed unsupervised methods in detection accuracy and recall.
- Neural networks showed good learning behavior but were slightly outperformed by XGBoost in generalization.
- The system is well-prepared for deployment with strong test results and explainability, though rare attack types remain a challenge.



RECOMMENDATIONS

- Enhance Class Balance.
- Expand Interpretability: Integrate SHAP for both global and local explanation perspectives
- Complete Deployment: Finalize real-time streaming using River, Kafka, or custom tools with .pcap integration
- Model Fusion: Experiment with ensemble models combining Neural Network and XGBoost to capture complex patterns
- Integrate Threat Response: Automate actions like IP blocking, alert generation, and network isolation based on prediction output

THANK YOU

Does anyone have any questions?



Contributors:
Julie Maina
Olin Wachira
David Mburu
Denzel Tero
Lyster Kwamboka